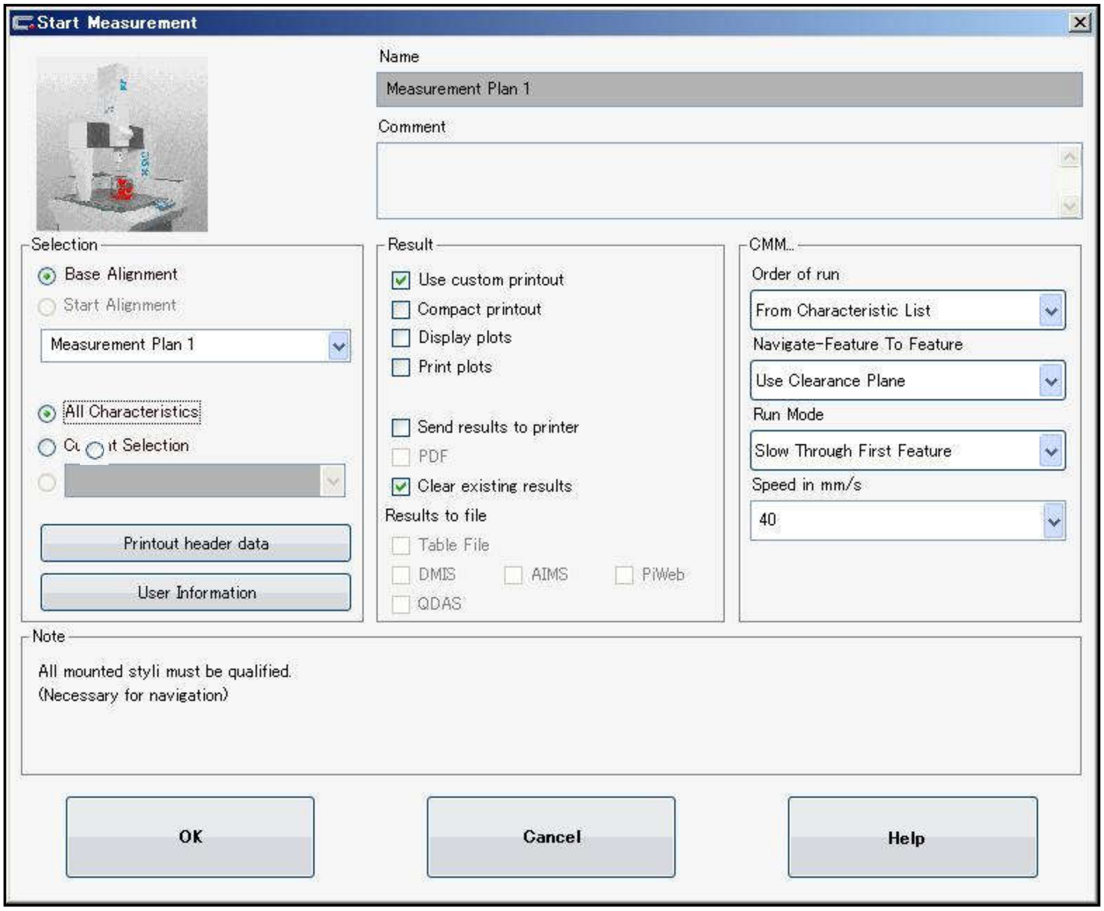
(6) Measurement plan execution

1. Click "CNC-Start" icon .

   Measurement Start window appears. Specify as follows.

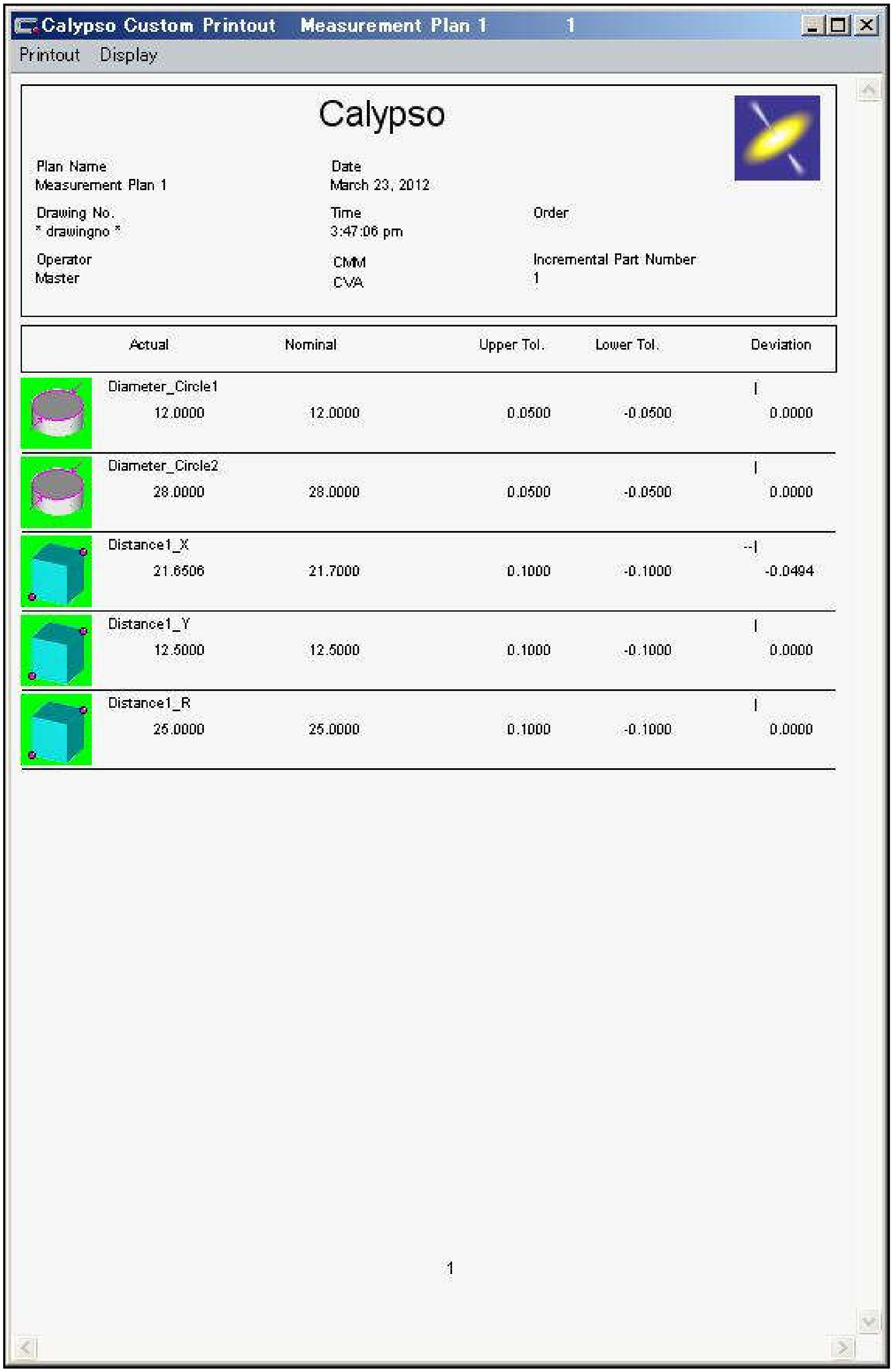


2. **<u>Move the stylus to the upper part of the work</u>.**
3. Click "OK" to start the measurement.

   The result shown as the next page is outputted after the measurement.

* For the Planes from 1 to 3, the measurement is proceeded while changing the stylus automatically.

Example of Custom Printout



<< Memo >>

Output of result

The result can be output in three types.

(1) Default printout

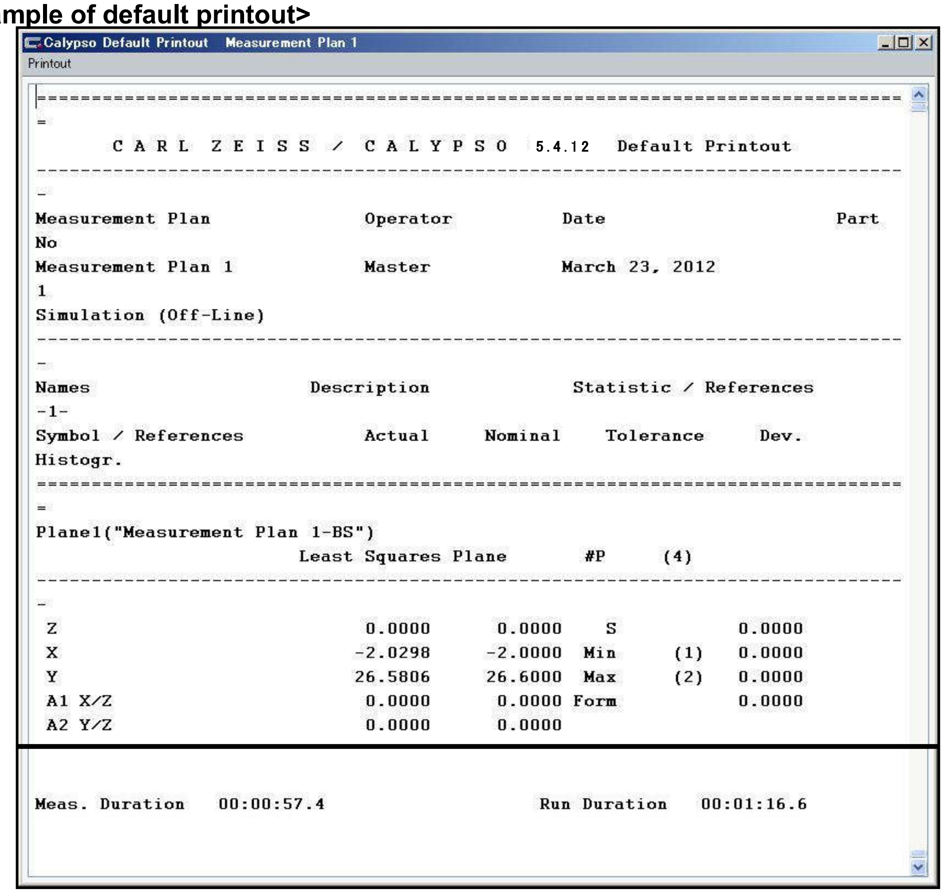
The display can be turned on or off. (P121 – Define printout)

By turning the display on, all the measurement results are always displayed on a separate window.

It can be used to check the progress of the measurement since the results are updated in real time even in the course of the automatic measurement.

Moreover, after a measurement is completed, it is possible to check the time required for the measurement.

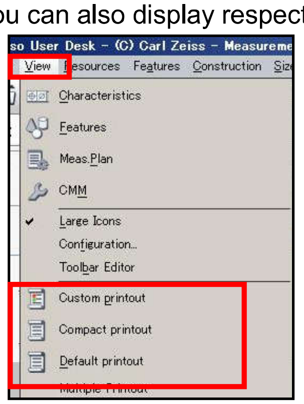
<Example of default printout>



```
Calypso Default Printout   Measurement Plan 1
Printout
=============================================================================
=
      C A R L  Z E I S S  /  C A L Y P S O  5.4.12  Default Printout
-----------------------------------------------------------------------------
-
Measurement Plan              Operator         Date                     Part
No
Measurement Plan 1            Master           March 23, 2012
1
Simulation (Off-Line)
-----------------------------------------------------------------------------
-
Names                    Description            Statistic / References
-1-
Symbol / References           Actual     Nominal    Tolerance     Dev.
Histogr.
=============================================================================
=
Plane1("Measurement Plan 1-BS")
                        Least Squares Plane       #P      (4)
-----------------------------------------------------------------------------
-
 Z                            0.0000      0.0000   S              0.0000
 X                           -2.0298     -2.0000  Min      (1)    0.0000
 Y                           26.5806     26.6000  Max      (2)    0.0000
 A1 X/Z                       0.0000      0.0000 Form             0.0000
 A2 Y/Z                       0.0000      0.0000

Meas. Duration    00:00:57.4                   Run Duration    00:01:16.6
```

* The system shall output the designated result upon completed auto measurement.

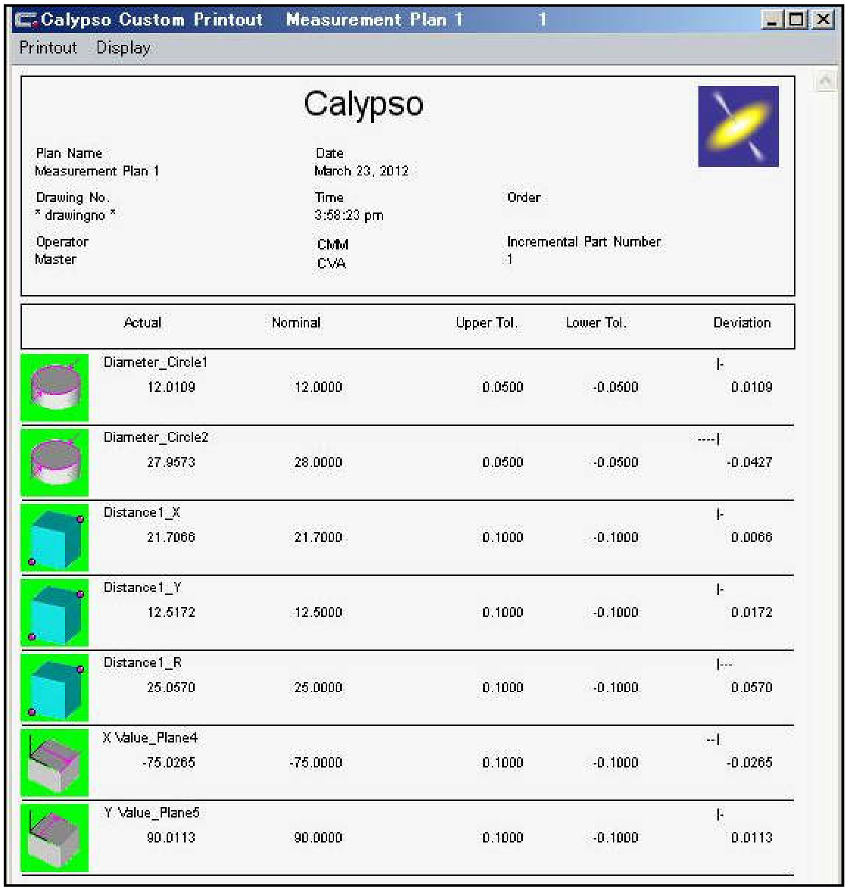
You can also display respective result item from the Menu bar – [Display].



(2) Custom printout

It is the standard output form of Calypso.
Not all the results but only the results registered in advance are displayed.
The output order can be rearranged as needed; emphasizing on the visibility.

<Example of custom printout> * Display by default



Calypso Custom Printout Measurement Plan 1 1

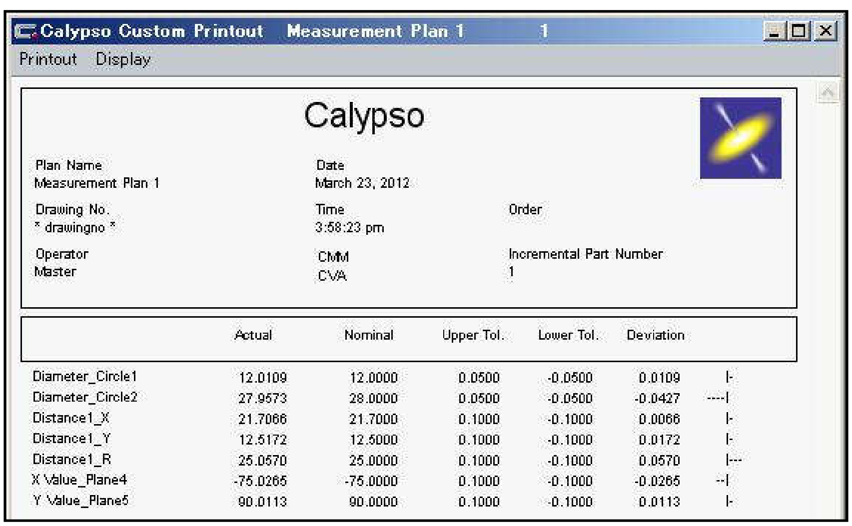
Printout Display

Calypso

Plan Name Measurement Plan 1	Date March 23, 2012	
Drawing No. * drawingno *	Time 3:58:23 pm	Order
Operator Master	CMM CVA	Incremental Part Number 1

	Actual	Nominal	Upper Tol.	Lower Tol.	Deviation
Diameter_Circle1					|-
	12.0109	12.0000	0.0500	-0.0500	0.0109
Diameter_Circle2					----|
	27.9573	28.0000	0.0500	-0.0500	-0.0427
Distance1_X					|-
	21.7066	21.7000	0.1000	-0.1000	0.0066
Distance1_Y					|-
	12.5172	12.5000	0.1000	-0.1000	0.0172
Distance1_R					|---
	25.0570	25.0000	0.1000	-0.1000	0.0570
X Value_Plane4					--|
	-75.0265	-75.0000	0.1000	-0.1000	-0.0265
Y Value_Plane5					|-
	90.0113	90.0000	0.1000	-0.1000	0.0113

<Example of custom printout>* Display not showing the icon (Printout format shown in P. 102)



Calypso Custom Printout Measurement Plan 1 1

Printout Display

Calypso

Plan Name Measurement Plan 1	Date March 23, 2012	
Drawing No. * drawingno *	Time 3:58:23 pm	Order
Operator Master	CMM CVA	Incremental Part Number 1

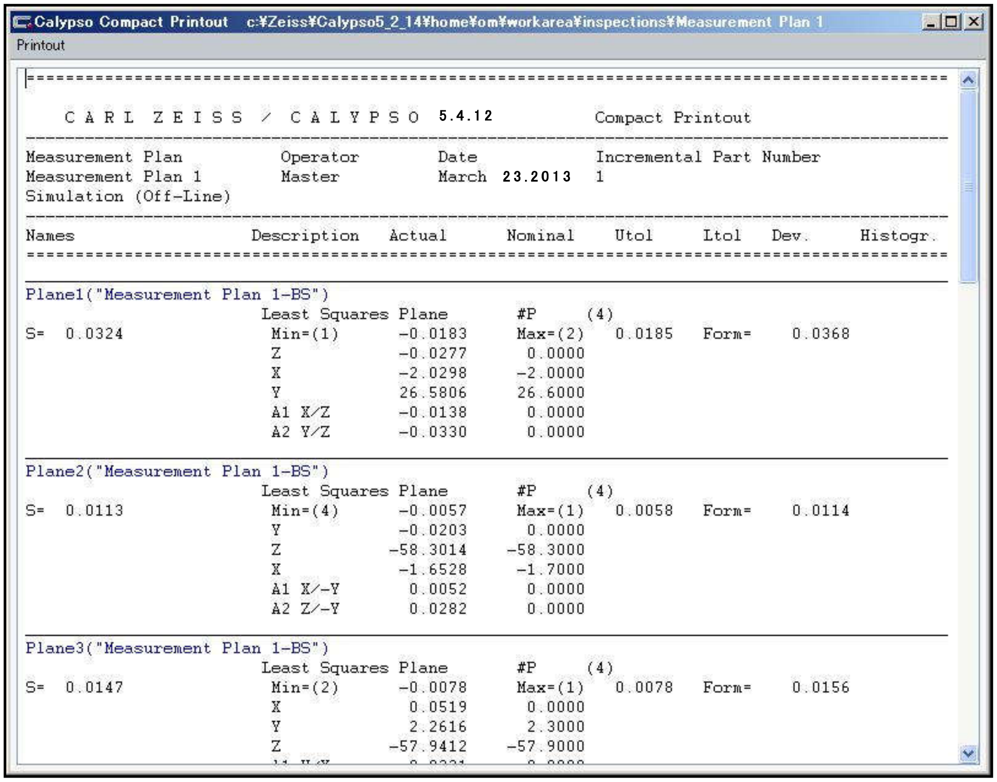
	Actual	Nominal	Upper Tol.	Lower Tol.	Deviation	
Diameter_Circle1	12.0109	12.0000	0.0500	-0.0500	0.0109	|-
Diameter_Circle2	27.9573	28.0000	0.0500	-0.0500	-0.0427	----|
Distance1_X	21.7066	21.7000	0.1000	-0.1000	0.0066	|-
Distance1_Y	12.5172	12.5000	0.1000	-0.1000	0.0172	|-
Distance1_R	25.0570	25.0000	0.1000	-0.1000	0.0570	|---
X Value_Plane4	-75.0265	-75.0000	0.1000	-0.1000	-0.0265	--|
Y Value_Plane5	90.0113	90.0000	0.1000	-0.1000	0.0113	|-

⑶ Compact printout

All the measured results are displayed.

The character spacing and gaps of the default printout are adjusted; more legible than the default printout.

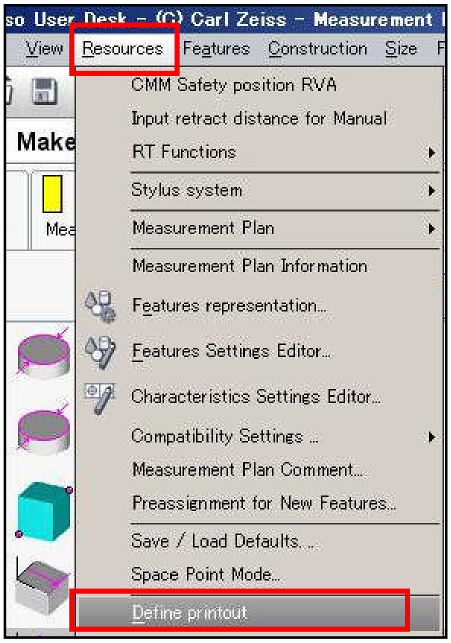
<Example of compact printout>



```
Calypso Compact Printout    c:¥Zeiss¥Calypso5_2_14¥home¥om¥workarea¥inspections¥Measurement Plan 1
Printout

==========================================================================================

    C A R L  Z E I S S  /  C A L Y P S O  5.4.12          Compact Printout
------------------------------------------------------------------------------------------
Measurement Plan          Operator        Date            Incremental Part Number
Measurement Plan 1        Master          March  23.2013  1
Simulation (Off-Line)
------------------------------------------------------------------------------------------
Names                  Description   Actual      Nominal    Utol     Ltol    Dev.     Histogr.
==========================================================================================

Plane1("Measurement Plan 1-BS")
                        Least Squares Plane      #P      (4)
S=  0.0324               Min=(1)      -0.0183    Max=(2)   0.0185   Form=    0.0368
                         Z            -0.0277     0.0000
                         X            -2.0298    -2.0000
                         Y            26.5806    26.6000
                         A1 X/Z       -0.0138     0.0000
                         A2 Y/Z       -0.0330     0.0000

Plane2("Measurement Plan 1-BS")
                        Least Squares Plane      #P      (4)
S=  0.0113               Min=(4)      -0.0057    Max=(1)   0.0058   Form=    0.0114
                         Y            -0.0203     0.0000
                         Z           -58.3014   -58.3000
                         X            -1.6528    -1.7000
                         A1 X/-Y       0.0052     0.0000
                         A2 Z/-Y       0.0282     0.0000

Plane3("Measurement Plan 1-BS")
                        Least Squares Plane      #P      (4)
S=  0.0147               Min=(2)      -0.0078    Max=(1)   0.0078   Form=    0.0156
                         X             0.0519     0.0000
                         Y             2.2616     2.3000
                         Z           -57.9412   -57.9000
```

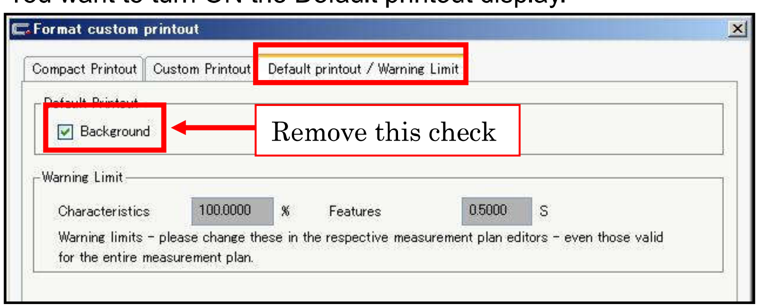
Reference: Printout format setting

①Select [Define printout] from the [Resources] menu.

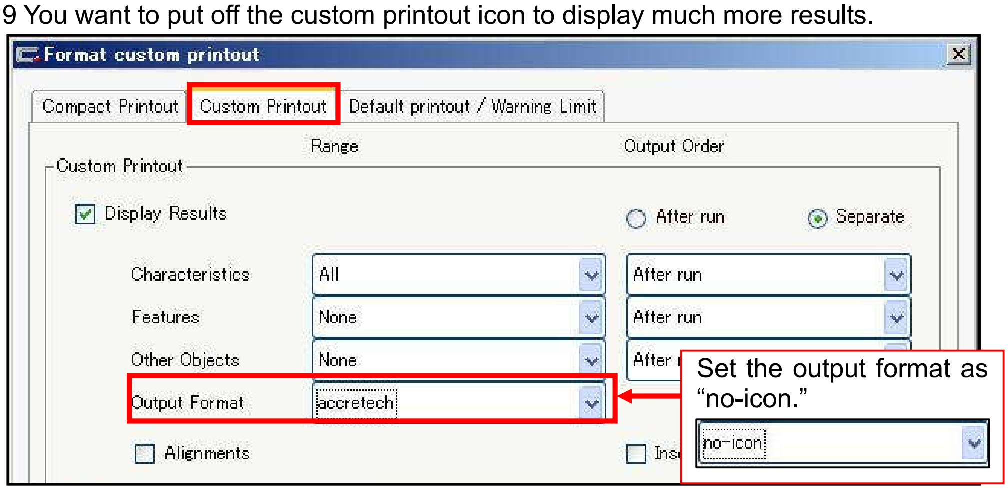


②As the following window appears, open a tab you want to set and do changing.

・P118 You want to turn ON the Default printout display.



・P119 You want to put off the custom printout icon to display much more results.

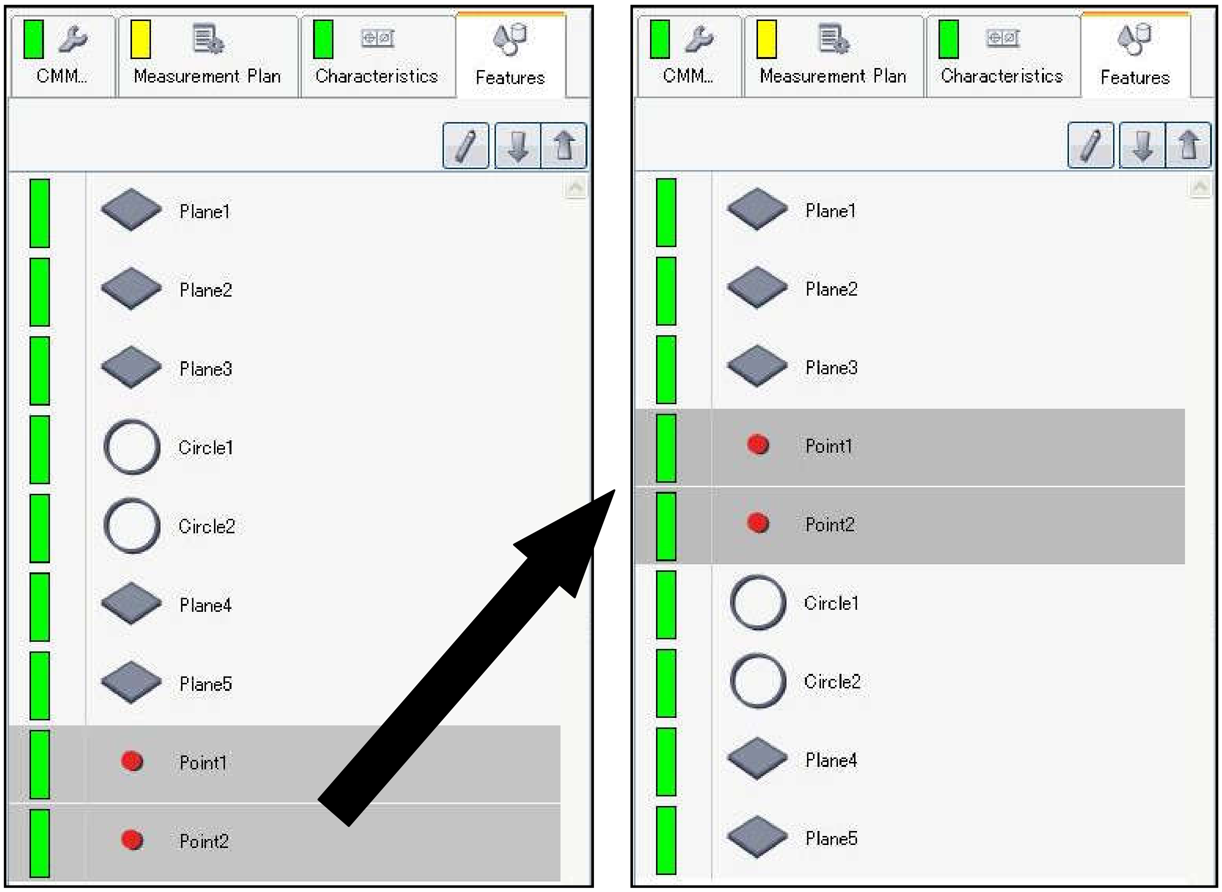


≪ Memo ≫

Edit of Measurement Plan

11-1. Change of orders for features and characteristics

It is possible to change the order of the measurement and output by changing the order of the list.
To change the order, place a mouse pointer on the feature or the characteristics icon intended.
Drag the icon to where you want to move while keeping the left button pressed, and release the left button.

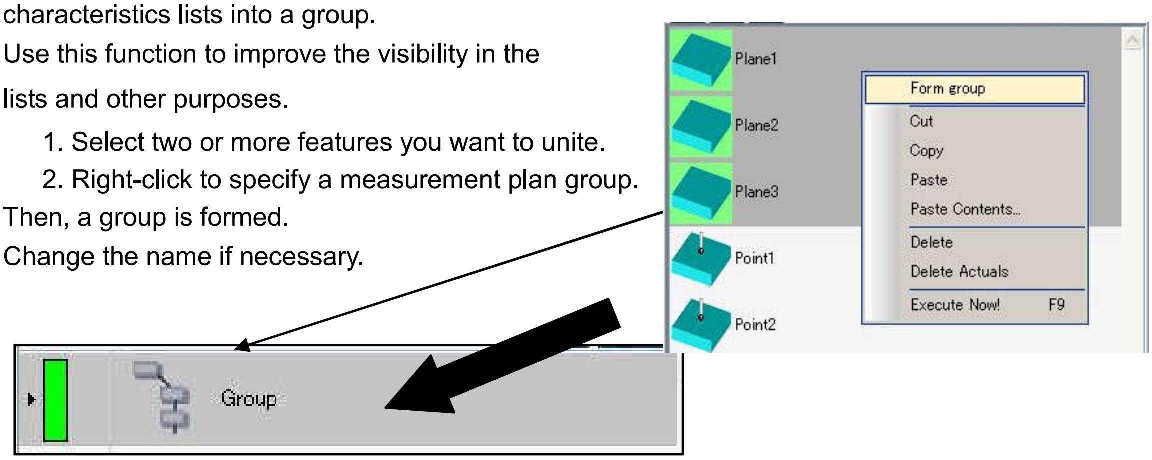


11-2. Grouping of Features and Characteristics

It is possible to unite the feature and the characteristics lists into a group.
Use this function to improve the visibility in the lists and other purposes.

1. Select two or more features you want to unite.
2. Right-click to specify a measurement plan group.

Then, a group is formed.
Change the name if necessary.

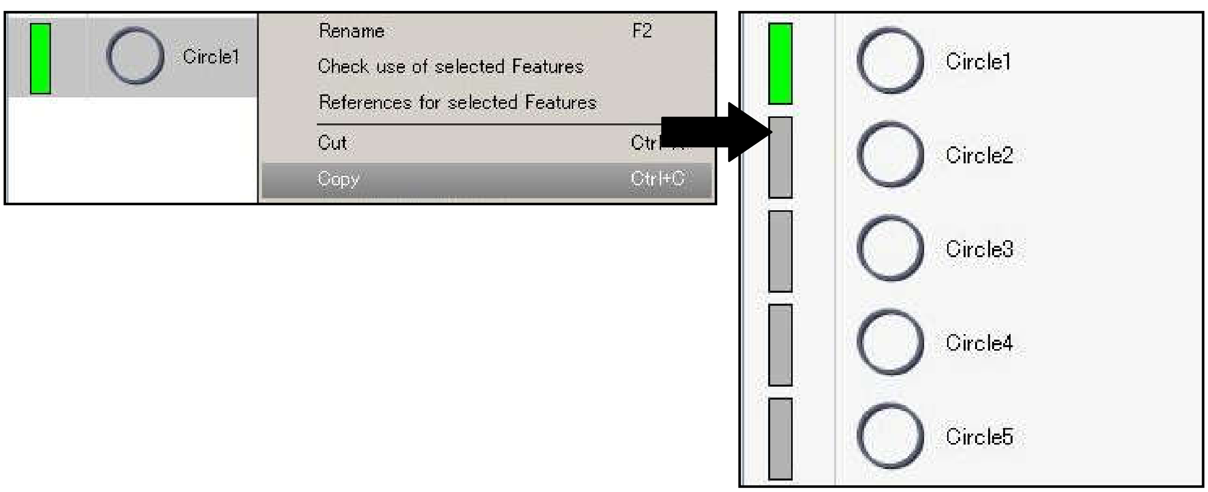


11-3. Renaming of Features and Characteristics

The name of a feature or a characteristic can be changed.
The renamed characteristics is outputted with the new name.
To change the name, right-click the target feature or characteristics, and click "Rename."

11-4. Copy & Paste

To make copies, the features and characteristics can be copied and pasted by right-clicking.



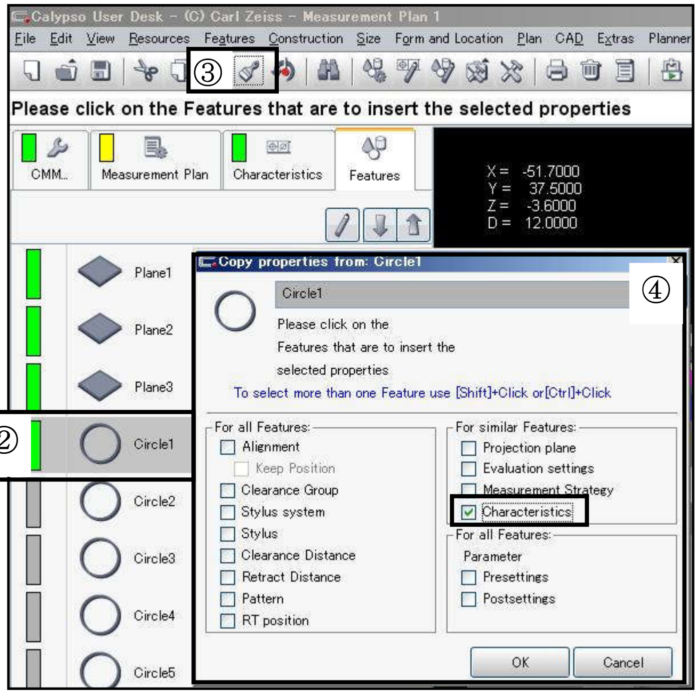
11-5.Copy properties

To the nomimal value and/or tolerance, or other similar elements of the charasteristic, you can copy a same charasteristic and allocate it.
For example, if you set the charasteristic X, Y and D to the circle 1 – 7, you might set only the circle 1 and use the Property copy to set the same task to the circle 2 – 7 all together.

① Set a task to the circle 1 and left click ON to close the feature window.

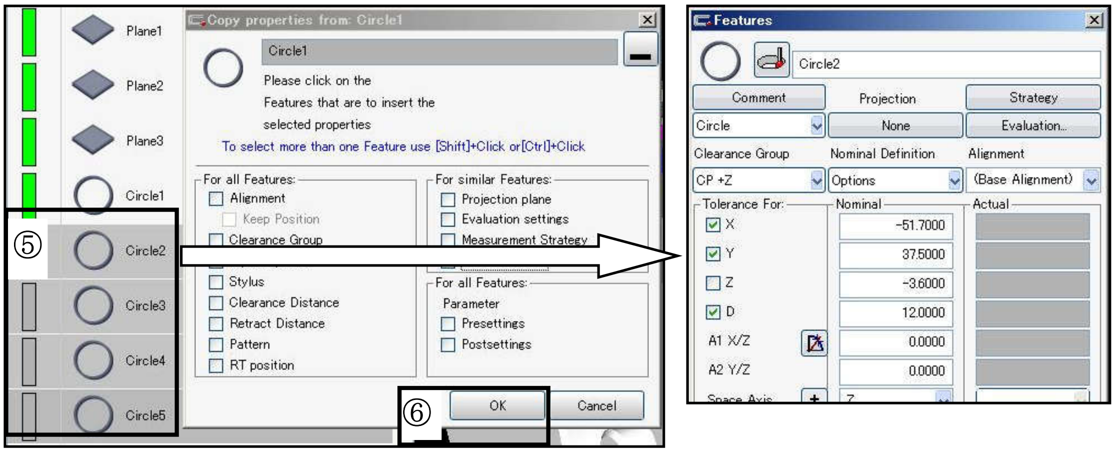
②Select the circle 1 (its color becomes back lighted).

③Left click the Copy properties icon to display the [Copy properties] window.



④Put a check in the item to copy.

⑤Keep opening the [Copy properties] window and select an element in the copy origin.



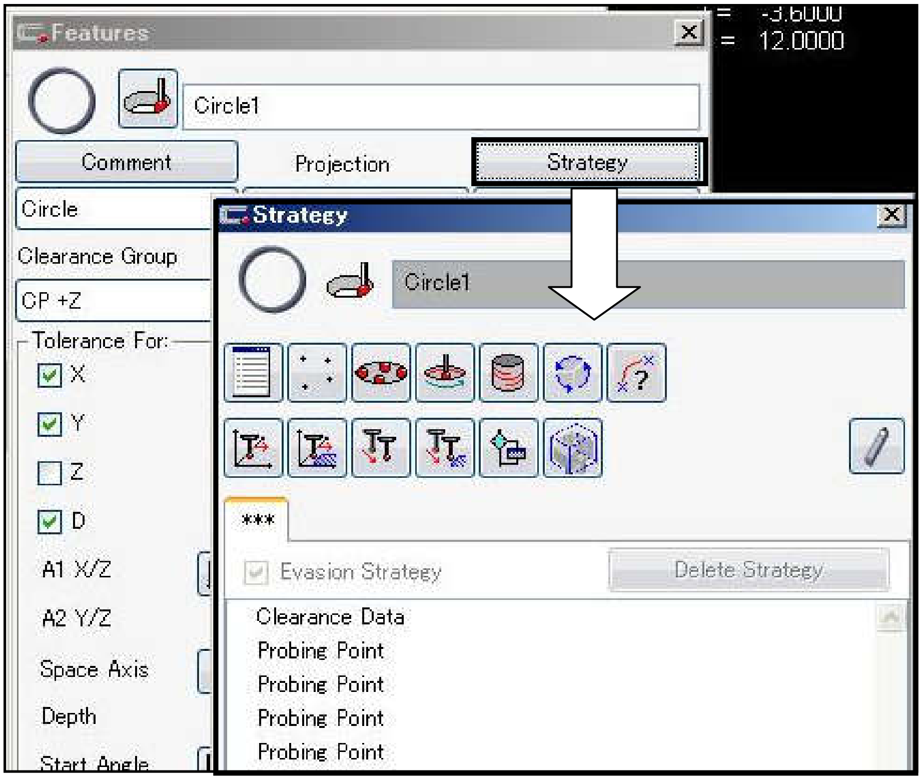
⑥Left click the OK button to close the [Copy properties] window.

11-6. Edit of measurement point

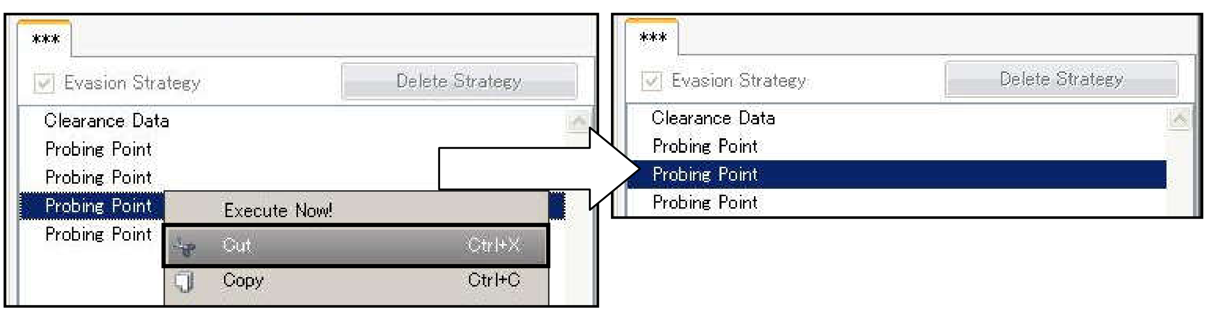
(1) Deletion of measurement point

If you don't want to measure (delete) the probing points or the half-way points already captured during CNC execution, perform the following operations.

1. Double-click the feature to edit.
2. Click "**Strategy**" button in the Features template to open Strategy window.



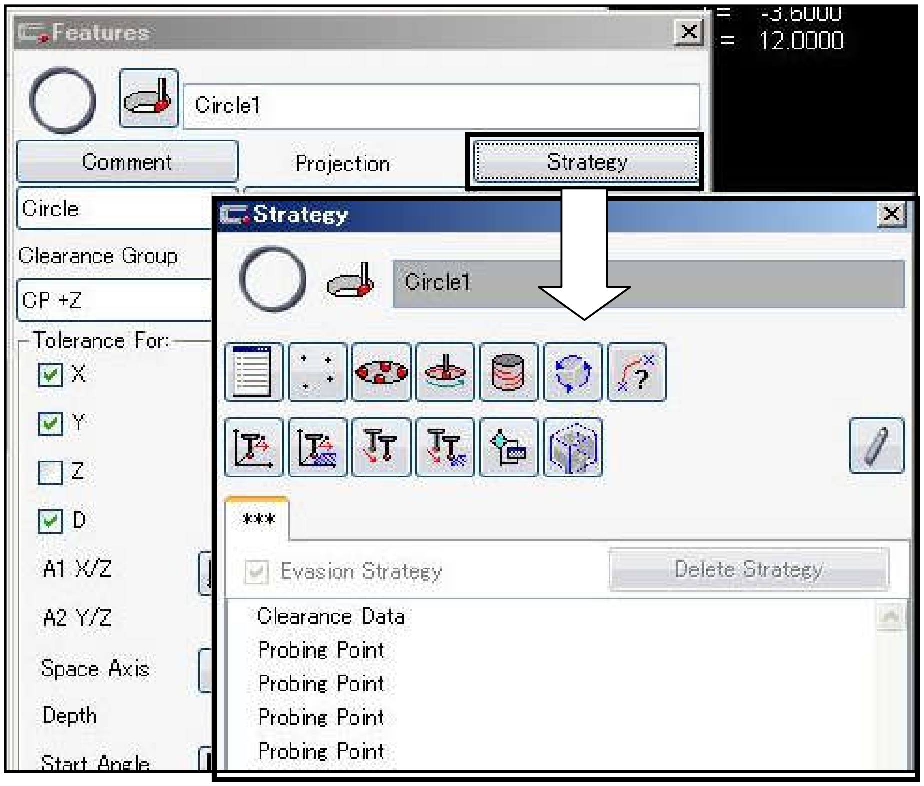
3. Click the item of the probing point (or half-way point) you want to delete, and make it highlighted.
4. Right-click and select "**Cut**" in the shortcut menu.



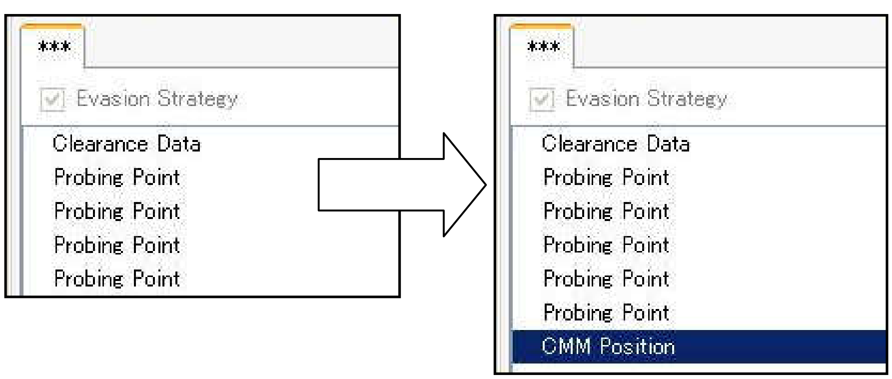
(2) Addition of measurement point

If you want to increase the probing points or half-way points to measure each feature, perform the following operations.

1. Double-click the feature to edit.
2. Click "**Strategy**" button in the Features template to open Strategy window.



3. With the Strategy window kept displayed, actually probe the feature.
   The "**Probing Points**" on display increase with each probing.
   To increase the half-way points, move the probe to the position intended, and press the top button on the right joystick.

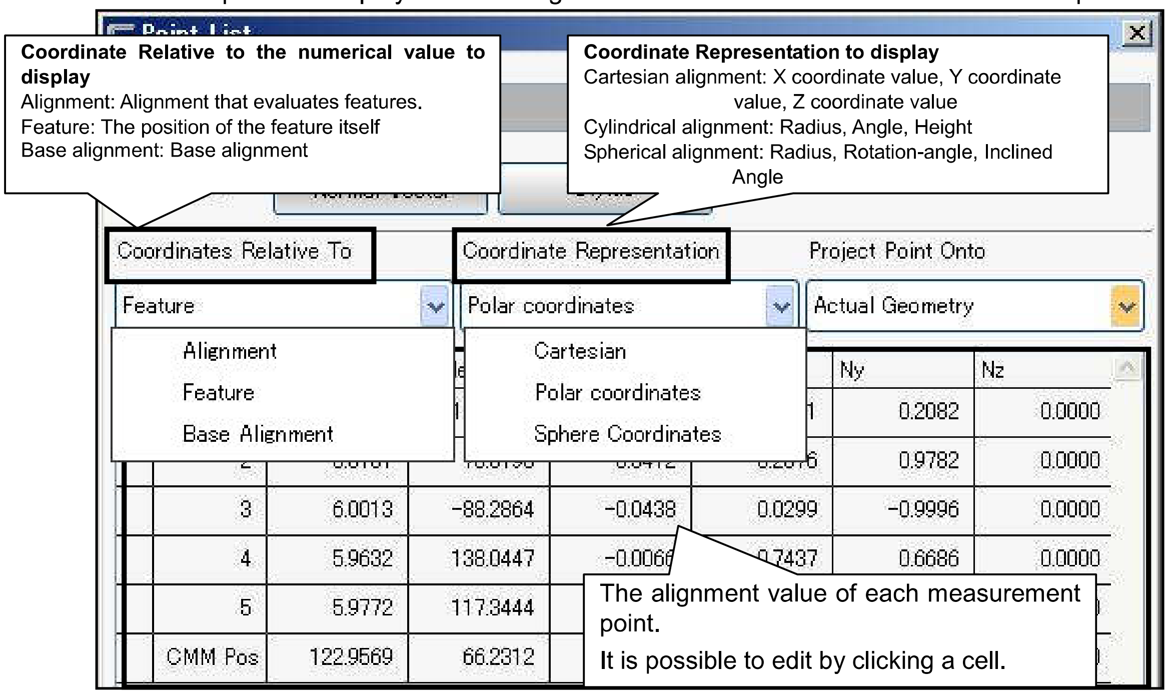


* You can change the order of added probing point and positioning of the Measuring machine (mid point) by drag.

(3) Edit of measurement point

The coordinate values such as probing and half-way points of measurement features can be edited individually by keyboard input.

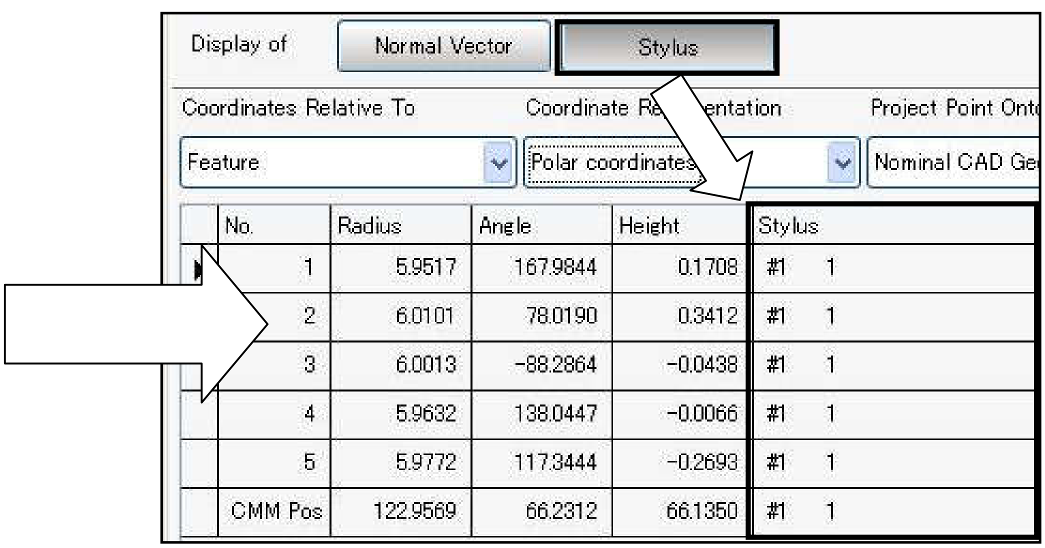
1. Double-click the feature to edit.
2. Click "**Strategy**" button in the Features template to open Strategy window.
3. Double-click a **probing point** from the list displayed on the Strategy window.
4. List window of points is displayed containing the coordinate value of each measurement point.



The list of points displays only the probing and half-way points.
If circle segments or others are specified, an individual setup is required.

*** For measuring one feature by two or more styluses**

Probing point / half-way point -> The setup details can be checked and changed on this screen.
Circle segment etc. -> The setup details can be checked and changed on respective setup window.



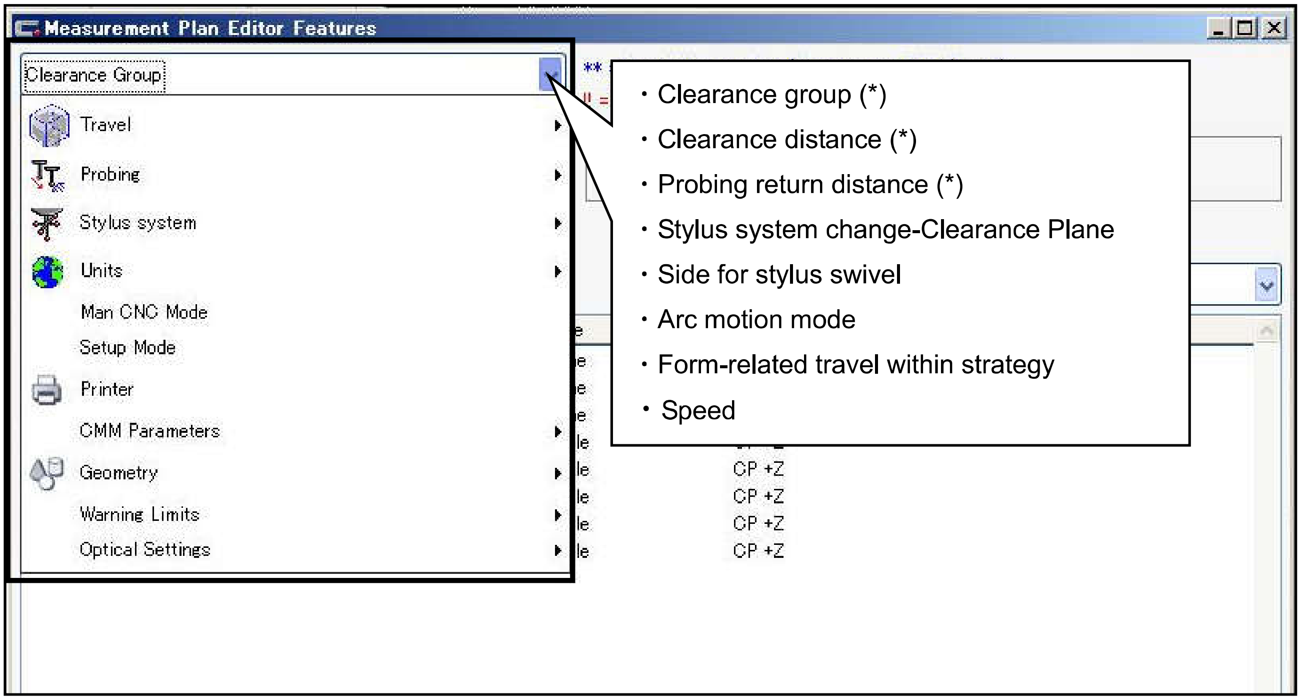
No.	Radius	Angle	Height	Stylus
1	5.9517	167.9844	0.1708	#1 1
2	6.0101	78.0190	0.3412	#1 1
3	6.0013	-88.2864	-0.0438	#1 1
4	5.9632	138.0447	-0.0066	#1 1
5	5.9772	117.3444	-0.2693	#1 1
CMM Pos	122.9569	66.2312	66.1350	#1 1

11-7 Measurement Plan Editor Feature

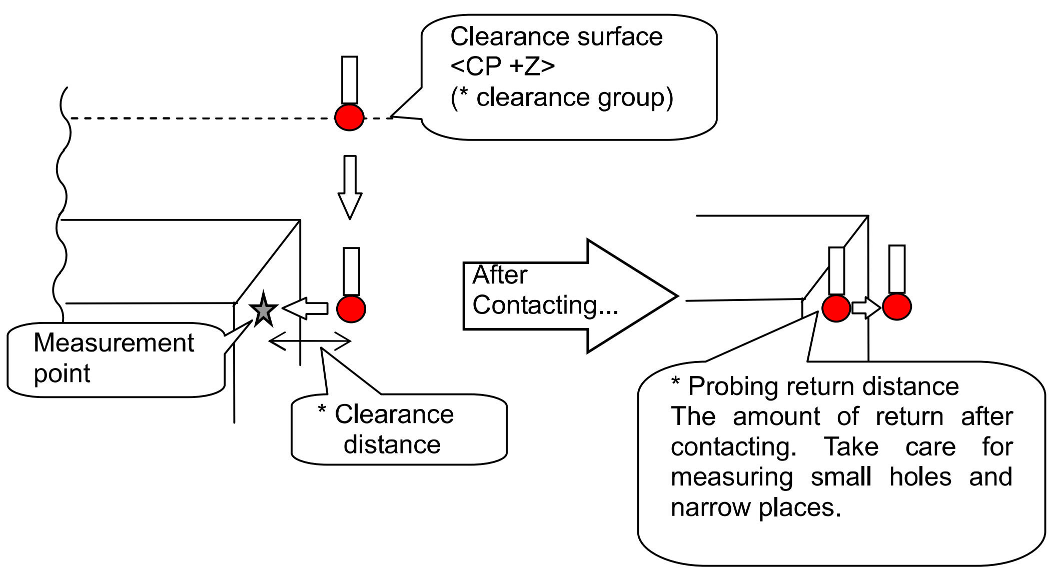
On the Measurement Plan Editor feature, the detailed setup on Measurement Plan can be done.

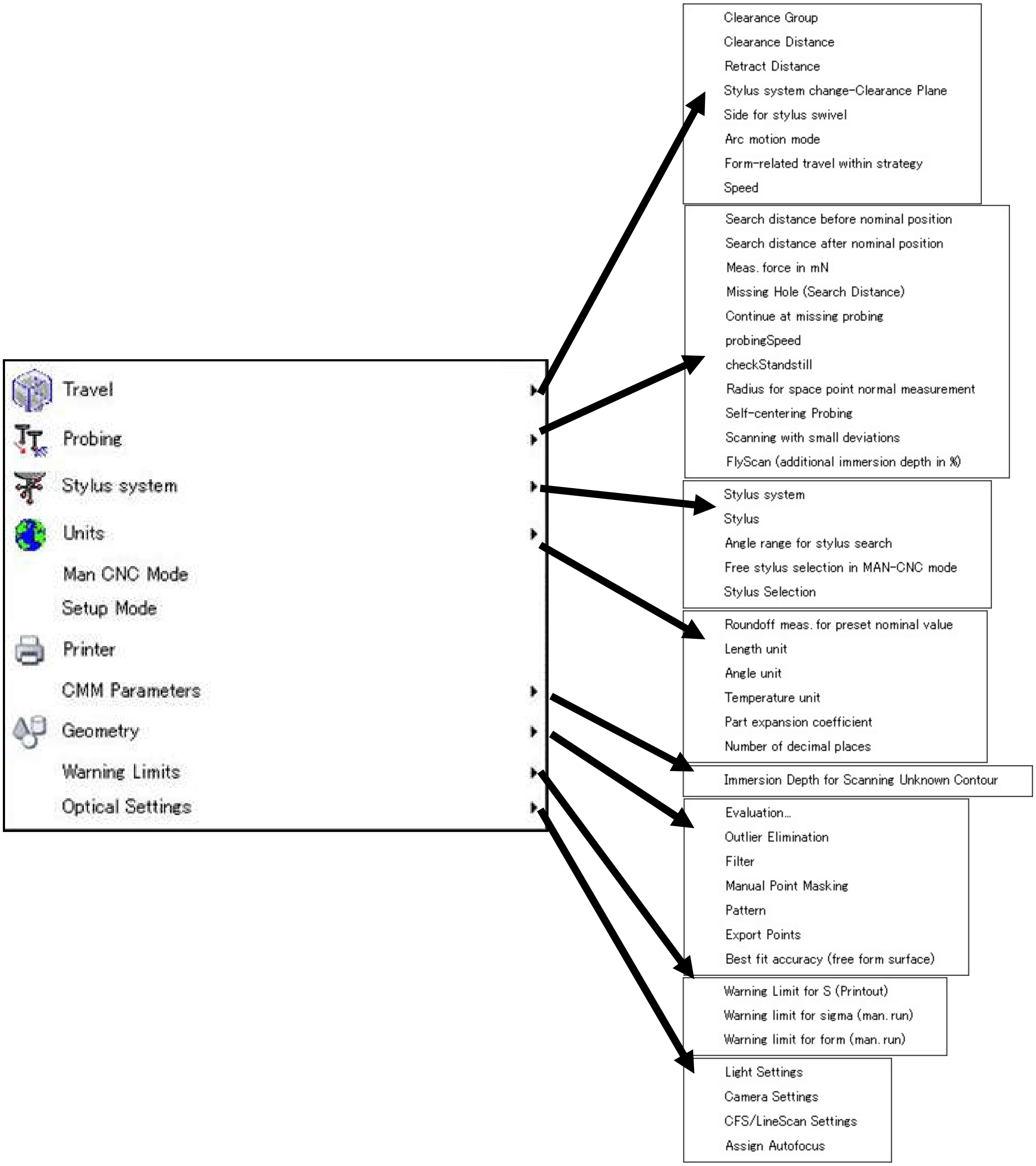
Click "Measurement Plan Editor Feature" icon in the Prerequisites List.

The setup items relating to movement during CNC mainly involve the **Clearance Group, Clearance Distance, and Retract Distance.**



Operation up to the measure point (for End Face)




Travel
Probing
Stylus system
Units
Man CNC Mode
Setup Mode
Printer
CMM Parameters
Geometry
Warning Limits
Optical Settings
Clearance Group
Clearance Distance
Retract Distance
Stylus system change-Clearance Plane
Side for stylus swivel
Arc motion mode
Form-related travel within strategy
Speed
Search distance before nominal position
Search distance after nominal position
Meas. force in mN
Missing Hole (Search Distance)
Continue at missing probing
probingSpeed
checkStandstill
Radius for space point normal measurement
Self-centering Probing
Scanning with small deviations
FlyScan (additional immersion depth in %)
Stylus system
Stylus
Angle range for stylus search
Free stylus selection in MAN-CNC mode
Stylus Selection
Roundoff meas. for preset nominal value
Length unit
Angle unit
Temperature unit
Part expansion coefficient
Number of decimal places
Immersion Depth for Scanning Unknown Contour
Evaluation...
Outlier Elimination
Filter
Manual Point Masking
Pattern
Export Points
Best fit accuracy (free form surface)
Warning Limit for S (Printout)
Warning limit for sigma (man. run)
Warning limit for form (man. run)
Light Settings
Camera Settings
CFS/LineScan Settings
Assign Autofocus

〈〈 Memo 〉〉

Auto Run